

Advance HR Analytics Dashboard using Power BI

Problem Statement

Organizations generate a large amount of employee data across recruitment, performance, salaries, satisfaction, and attrition. However, without an interactive reporting solution, HR teams struggle to identify workforce trends, monitor employee performance, evaluate recruitment effectiveness, and make informed business decisions.

This project aims to transform raw HR data into an interactive analytics solution that provides actionable insights for workforce planning and strategic HR decision-making.

Business Objective

Develop a comprehensive HR Analytics Dashboard that enables HR managers to:

- Monitor workforce demographics
 - Analyze employee attrition
 - Track recruitment effectiveness
 - Evaluate employee performance
 - Understand salary distribution
 - Measure employee satisfaction
 - Support data-driven HR decisions
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Key Dashboard Pages

1 Overview Dashboard

Provides a high-level summary of the organization's workforce.

KPIs

- Total Headcount
- Attrition
- Gender Distribution
- Average Employee Age

Visuals

- Department Headcount
- Hiring Trend
- Attrition Trend
- Recruitment Sources
- Age Distribution
- Marital Status Analysis

Business Questions Answered

- Which department has the largest workforce?
 - How has hiring changed over time?
 - What are the primary recruitment sources?
 - What is the current attrition rate?
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2 Employee Overview

Focuses on employee performance, compensation, and satisfaction.

KPIs

- Average Salary
- Projects Completed
- Average Tenure
- Satisfaction Score
- Yearly Absences

Visuals

- Salary by Department
- Termination Analysis
- Performance Distribution
- Salary by Race

- Employment Status
- Department Satisfaction

Business Questions Answered

- Which department has the highest salaries?
 - How satisfied are employees?
 - What are the major reasons for employee termination?
 - Which departments have the strongest performers?
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Workforce Analytics

Provides workforce planning insights.

KPIs

- Active Employees
- Current Year Hires
- Previous Year Hires
- Hiring Growth
- Average Tenure

Visuals

- Gender Distribution by Department
- Recruitment Funnel
- Employee Performance Table
- Age Matrix

Business Questions Answered

- Which recruitment source performs best?
- Which department hires the most employees?
- How is the workforce distributed across age groups?
- What is the hiring trend over time?

Employee 360°

Interactive Drill-through Page

Displays employee-level information including:

- Salary
- Department
- Position
- Performance
- Satisfaction
- Tenure
- State
- Marital Status
- Special Projects

Allows HR managers to investigate individual employee records directly from the dashboard.

Skills Demonstrated

Power BI

- Data Modeling
- Star Schema
- Relationships
- Drill Through
- Drill Down
- Bookmarks
- Page Navigation
- Interactive Filtering

- Conditional Formatting
 - Custom Tooltips
 - KPI Cards
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Power Query

- Data Cleaning
 - Data Type Conversion
 - Remove Columns
 - Replace Values
 - Split Columns
 - Conditional Columns
 - Group By
 - Data Transformation
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DAX

Implemented advanced DAX measures using:

- CALCULATE()
- FILTER()
- DIVIDE()
- COUNT()
- DISTINCTCOUNT()
- AVERAGE()
- IF()
- SWITCH()
- MAX()
- MIN()

- Time Intelligence concepts for hiring trend analysis
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Dashboard Features

- ✓ Interactive Navigation
 - ✓ Dynamic Slicers
 - ✓ Drill-through
 - ✓ Conditional Formatting
 - ✓ Business KPIs
 - ✓ Matrix Visualization
 - ✓ Funnel Chart
 - ✓ Treemap
 - ✓ KPI Cards
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Business Insights

Some important insights generated from the dashboard include:

- Production department has the largest workforce.
- Employee attrition is concentrated during specific hiring years.
- LinkedIn and Indeed contribute the highest number of successful hires.
- Most employees belong to the 36–45 age group.
- Executive Office receives the highest average salary.
- Software Engineering department records the highest employee satisfaction.
- "Fully Meets" is the most common performance rating.
- The majority of employees remain in Active employment status.

Recommendations

- Improve employee retention initiatives in departments with higher attrition.
- Increase investment in high-performing recruitment channels such as LinkedIn and Indeed.
- Monitor employee satisfaction regularly through HR engagement programs.
- Focus leadership development initiatives on high-performing employees.
- Use Employee 360° reports for performance reviews and succession planning.